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Q: arr = [2, 3, 5, 9, 14, 16, 18], target = 14

Ceiling = smallest element in array greater or = target.

$$\text{Ceiling}(\text{arr}, \text{target} = 14) = 14$$

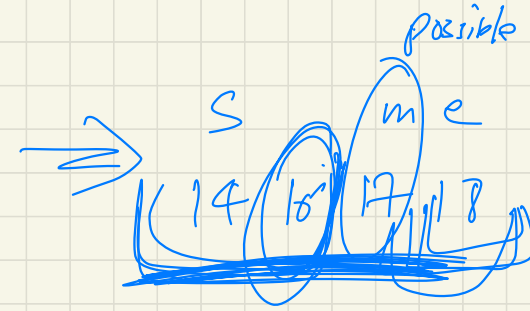
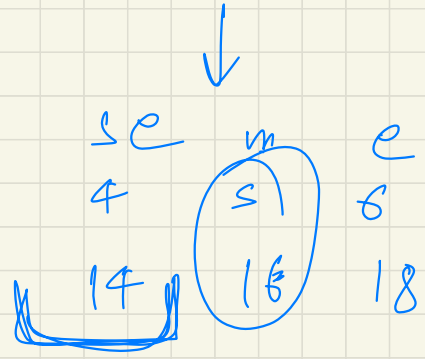
$$\text{Ceiling}(\text{arr}, \text{target} = 15) = 16$$

$$\text{target} = 4 = 5$$

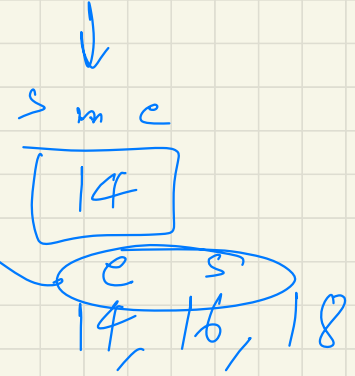
$$\text{target} = 9 = \underline{\underline{9}}$$

arr = [2, 3, 5, 9, 14, 16, 18]

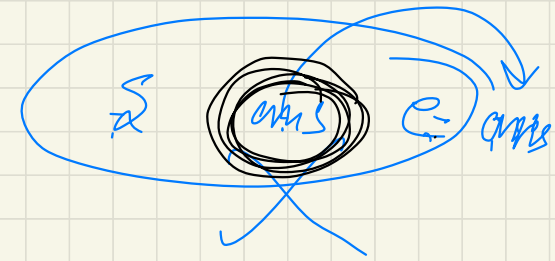
target = 15

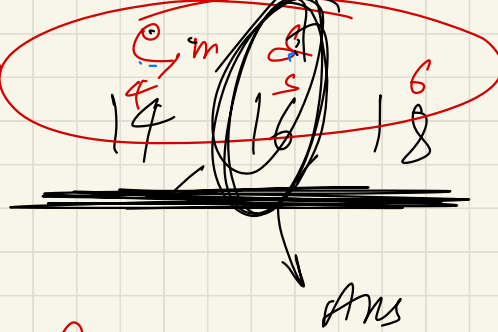


breaking  
loop  
while (s <= e)



target > mid  
⇒ s = m + 1





let's talk about  
start & end pointer.

Q: Find a num  
 $\geq$  target

$s$  (target)  $e$   
 $\Downarrow$

$e$  (target)  $s$  // condition for while loop violated

$start \leq end$

$\Rightarrow$  when while loop breaks  
 $start = end + 1$

next big no, when no ans found = start element. reason?

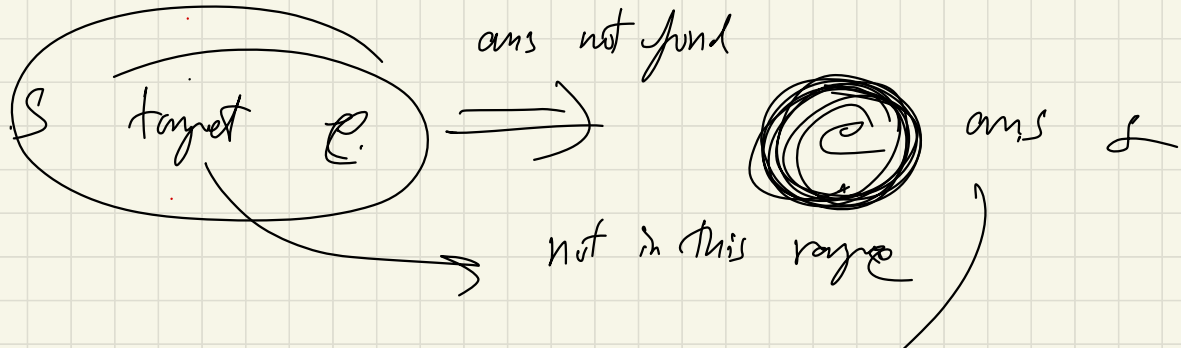
Q: Find the floor of a number.

floor = greatest number smaller or = target

arr = [2, 3, 5, 9, 14, 16, 18]      floor of 15

Ans = 14

// same thing as before, just return end instead of -1, if no. not found.



When ~~condition~~ it violated,

$$l = e + 1$$

$$s > e$$

Q3: (1) Exact same approach for finding of no.

(2) Ignore the target = what we are looking for here

(3) arr = ['c', 'd', 'y', 'j'], target = 'j'

Condition violated:  $start = end + 1 \Rightarrow$  length of array = N

return

$S \% N$

|

~~$2 \% 4 = 2$~~  // ret  
start

$4 \% 4 = 0$

if  $S == N$  :

return 0

Q4 :

arr = [ <sup>0</sup>5, <sup>1</sup>7, <sup>2</sup>7, <sup>3</sup>7, <sup>4</sup>7, <sup>5</sup>8, <sup>6</sup>8, <sup>7</sup>10 ]

target = 7

★ Find first occurrence of 7

[1, 4]

$e = m - 1$

Ans  
8

$e$   $s$   $m$   $e$   
[ 5, 7, 7, 7, 8, 8, 10 ]

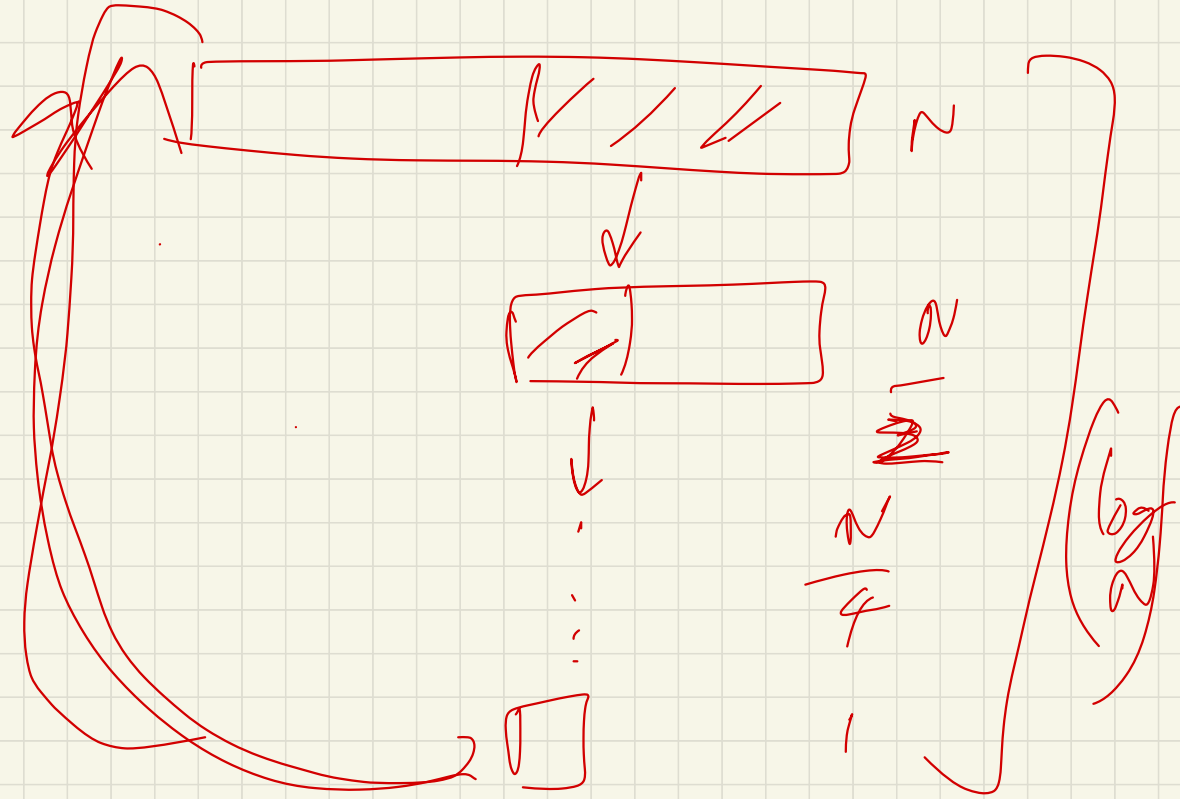
last occurrence :

$s = m + 1$

arr = [2, 3, 5, 6, 7, 8, 10, 11, 12, 15, 20, 23, 30]

target = 15

$\log N$



$5, 2, 3, 5, 6, 7, 8, 10, 11, 12, 15, 20, 23, 30$   
 (Note: The number 8 is circled in black, and 10 is underlined in red. There are also some blue and red markings above the numbers.)

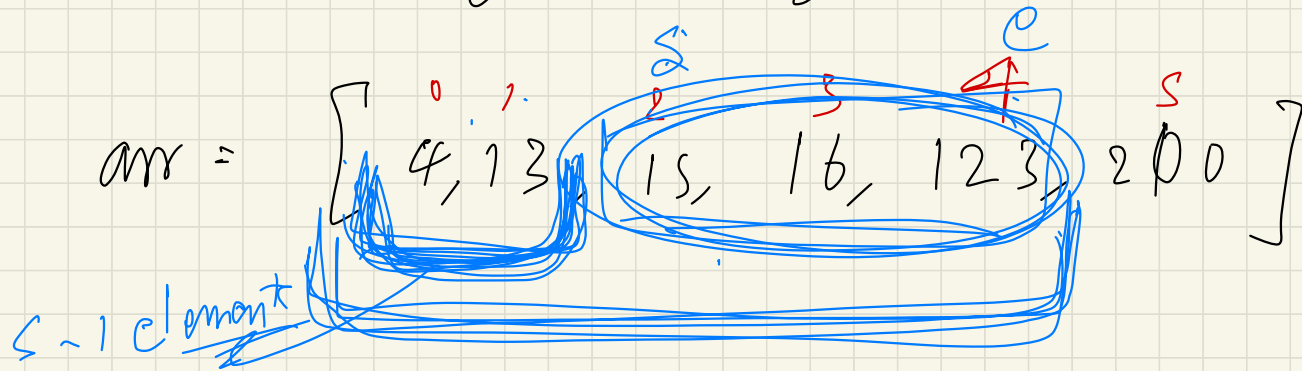
$$1 + 4 = 5$$

(Note: A red arrow points from the circled 8 to the text "any BS", which is underlined in red.)

$$5 + 2 \times 4 = 8 + 5 = 13$$

$$\begin{aligned}
 4 &= e - s + 1 \\
 &= 5 - 2 + 1 = 4
 \end{aligned}$$

Finding size of array by indices:

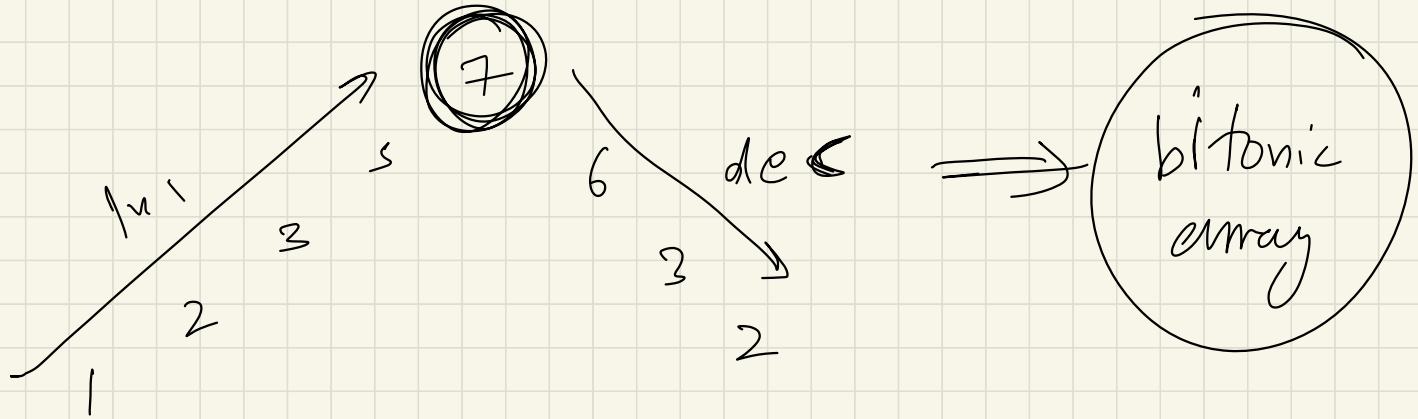


$$e - (s - 1) = (e - s + 1)$$

Ans

$$4 - 2 + 1 = 3$$

arr = [1, 2, 3, 5, 7, 6, 3, 2]



Q:

Find peak in mountain array.

① → Ans  
0 0

arr = [0, 2, 1, 0]

② → Ans index = 1  
0 1 0

arr = [1, 2, 3, 5, 6, 4, 3, 2]

Indices: 0 1 2 3 4 5 6 7

Annotations: A red line connects the first '1' to the first '5'. A blue line connects the first '5' to the first '3'. Both '5' and '3' are circled in blue and labeled 'm'. A red '2' is under the first '1', and a red '2' is under the first '3'.

①

if  $e[mid] > e[mid+1] \Rightarrow$  You are in the dec part of array

$\Rightarrow e = mid$   
// check in l.h

$s \left[ \begin{matrix} m \\ e \end{matrix} \right]$

②

$e[mid] < e[mid+1] \Rightarrow$  You are in asc part of arr.

$\Rightarrow e = mid + 1$

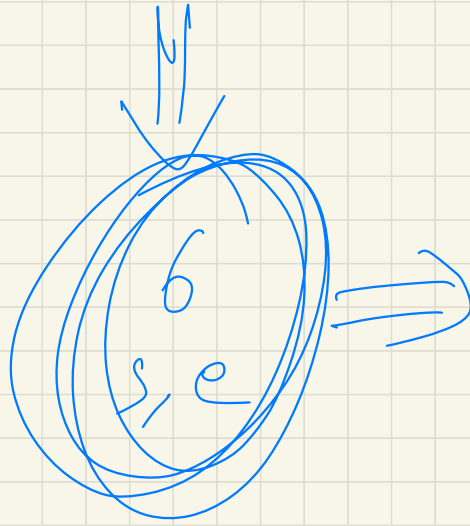
$\Rightarrow s \left[ \begin{matrix} m \\ mid+1 \\ e \end{matrix} \right]$

③ when will loop break?

0 1 2 3 4 5 6 7  
1, 2, 3, 5, 6, 4, 3, 2  
s m e

6, 4, 3, 2  
s m e

4 s  
6, 4  
s e  
m



In the end,  
s & e both  
will pt to  
the largest  
no.

Ans

why? ① & ②!

Q:

arr = [ 1, 2, 3, 4, 5, 3, 1 ]

target = 3

Indices: 0 1 2 3 4 5 6

Ans:

- ① Find peak element  $\Rightarrow$  4 index
- ② <sup>Binary</sup> Search in arr  $\Rightarrow$  (0, 4)
- ③ If not found, binary search in  $[4, 6]$  arr

Qr Rotated Binary Search:

arr = [2, 4, 5, 7, 8, 9, 10, 12]

After 1 rotation:

= [12, 2, 4, 5, 7, 8, 9, 10]

1 rotation

2<sup>nd</sup> rotation

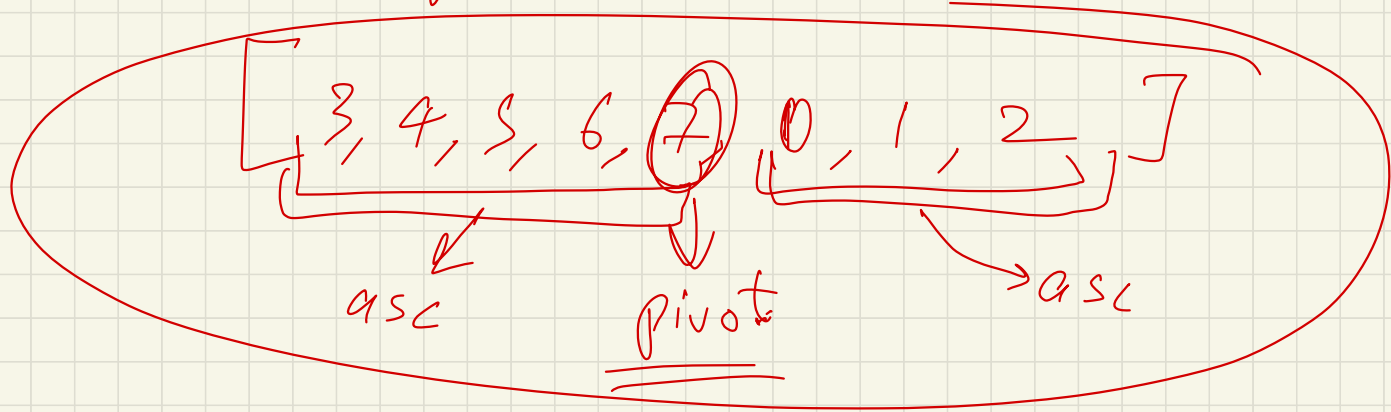
⇓

= [10, 12, 2, 4, 5, 7, 8, 9]

↘ pivot

①  $\Rightarrow$  Find the pivot in the array.

pivot  $\Rightarrow$  from where your next nos are asc.



★ Find pivot

★ Search in first half  $\Rightarrow$  Simple BS.  
(0, pivot)

★ otherwise, search in second half:  $(p+1, \text{end})$

Q: find pivot:

$$\begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ [3, & 4, & 5, & 6, & 7, & 0, & 1, & 2] \end{matrix}$$
  
 $\underbrace{\quad}_s \quad \quad \quad \underbrace{\quad}_m \quad \quad \quad \underbrace{\quad}_e$

only these 2 will be

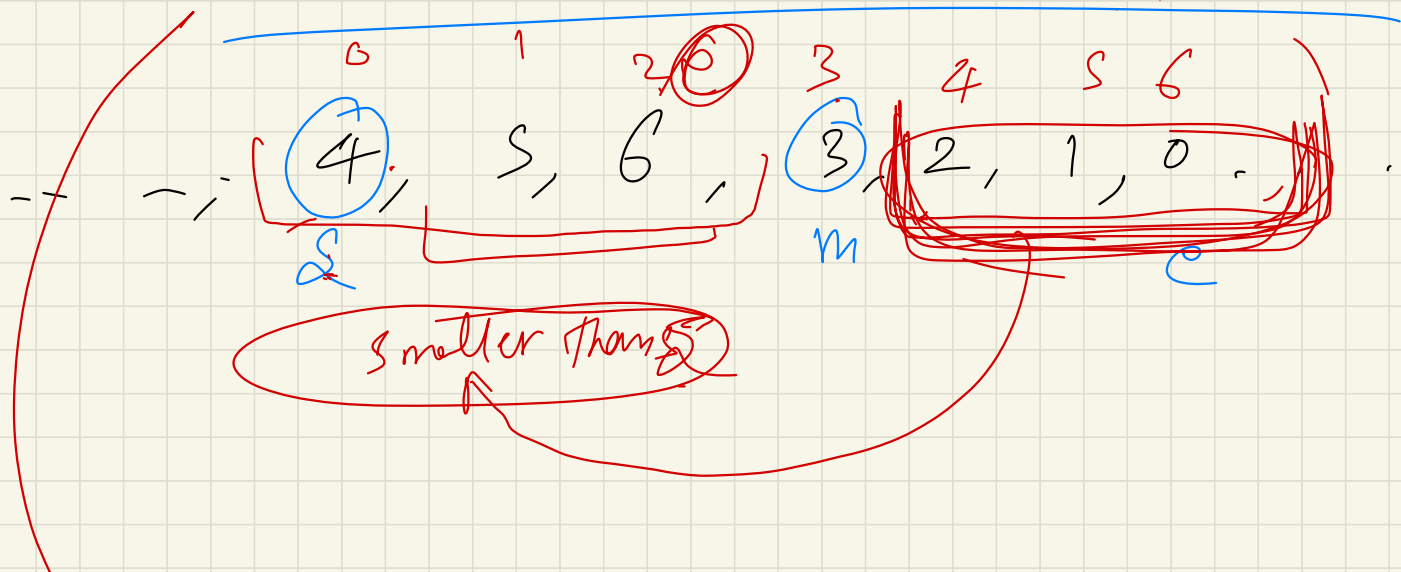
Ans when?  $\rightarrow$  when you find that  $\text{mid} > \text{mid} + 1$   
element, i.e. pivot. // case 1

Case 2 if

mid element  $<$  (mid-1) element  
i.e also my ans  $\Rightarrow$  ans =  $n-1$

Case 3

start element  $>$  mid-element



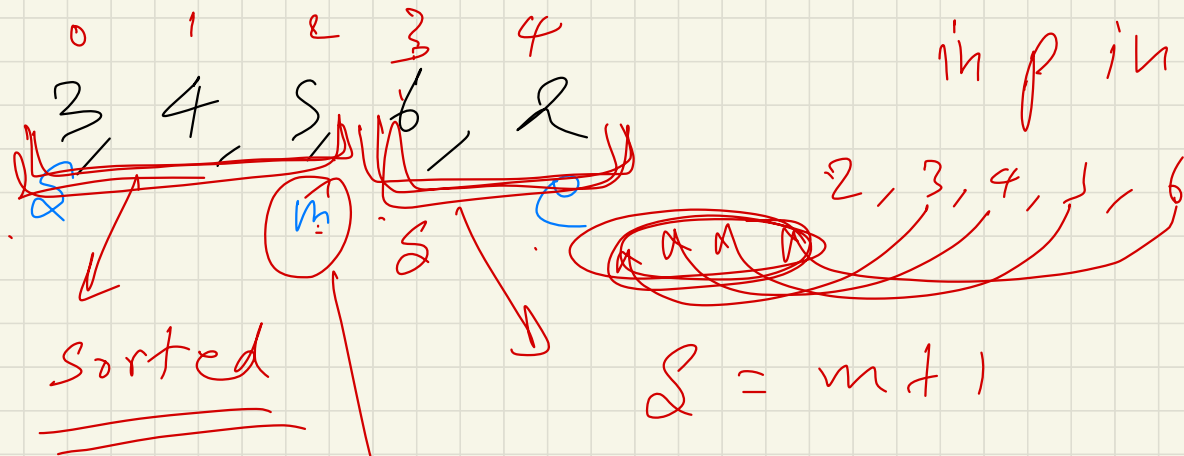
→ In this case, all elements from mid, will be  $<$  start.

Hence, we can ignore all these elements since we are looking for peak i.e. largest element.

$$e = \text{mid} - 1$$

Case 4:

$S$  element  $<$  mid element



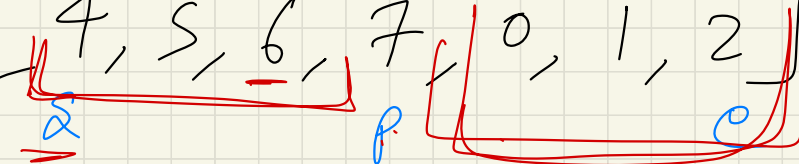
If this way pivot,  
it would've been  
returned in case 2.

Have proved, that bigger nos.  lie ahead.  
Hence, ignore mid & put  $s = mid + 1$

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Do RBS using pivot.

arr = [4, 5, 6, 7, 0, 1, 2]



Case 1 : pivot element = target // ans

Case 2 : target > start element // search for 6  
search space =  $(s, p-1)$

why? cuz all nos. after pivot  
are <  $s$

Case 3 : target < start element

i.e. we know that all elements  
from  $s$ , pivot are too going to be

bigger than target. // target = 1

Search space = (pivot + 1 till end)

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arr = [2, 2, 2, 2, 9]

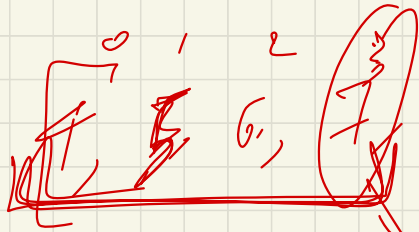
rotate twice

<sup>0</sup>  
[2, <sup>1</sup>9, <sup>2</sup>2, <sup>3</sup>2, <sup>4</sup>2]  
<sub>2</sub> <sub>arr</sub> <sub>e</sub>

arr = [4, 5, 6, 7, 0, 1, 2]



Rotated 4 times from original one



[0, 1, 2, 3]

~~4~~, ~~5~~, ~~6~~, ~~7~~

pivot = 3

Ans = 811

arr = [7, 2, 5, 10, 8], m = 2

7, 2, 5, 10  
24

8  
8

$\Rightarrow$

largest  
24

7, 2, 5  
14

10, 8  
18

18  
Ans

7, 2  
9

5, 10, 8  
23

23

7  
7

5, 10, 8  
23

25

★ (1) min n. of partitions that we can make = 1

★ (2) what is the knax n. of partitions / m  
that can be = N

arr = [3, 4, 1, 2]  $\Rightarrow$  [3], [4], [1], [2]

what will be the ans in case (1) :

[7, 2, 5, 10, 8]

Sum of entire array,

= 32

In Case (2) :

Ans for this = max element in  
array = 4

max value of ans of question = case 1

min value of ans of question = case 2

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minAns = max value in array

maxAns = sum of all values in array.

$$\text{start} = 10$$

$$\boxed{[10, 32]}$$
$$\text{end} = 32$$

Here it directed  
that we use  
BS.

$$\text{mid} = \frac{\text{start}}{2} = \frac{42}{2} = 21$$

Try to see if you can split the array with 21  
or the max sum:

7, 2, 5, 8, 10

$[7, 2, 5], [8, 10]$

pieces  
~~1~~ 2

check 1

if (pieces  $\leq$  m)

$\Rightarrow$  end = mid

2 = 10 , end = 21

mid = 15

7, 2, 5, 8, 10

[7, 2, 5], [8], [10]

pieces = 3

check 2

$\Rightarrow$  start = mid + 1

if pieces  $>$  m

$$s = 16, \quad e = 21$$

$$m = 18$$

$$7, 2, 5, 8, 10$$

$$[7, 2, 5], [8, 10]$$

$$\underline{\underline{\text{pieces} = 2}}$$

$$\Rightarrow s = 11, \quad e = 18$$

$$m = 17$$

$$7, 2, 5, 8, 10$$

$$[7, 2, 5], [8], [17]$$

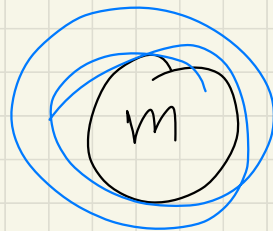
$$\underline{\underline{\text{pieces} = 3}}$$

$$Q = m + 1 = 18$$

$$S = 18, \quad e = 18$$

$$m = 18$$

Ans



// The ans exists definitely, hence by the above 2 checks we will reach it

